

Shell Scripting is Still Going Strong

This article introduces you to the basics of shell scripting and its importance in day-to-day life. A shell script is a command-line interpreter that runs on a UNIX/Linux shell.



The first thing we notice when we log into a UNIX/Linux system is the blinking cursor next to the \$ sign. This is the shell. It has been – for many decades – the ubiquitous (and many times the only) interface to interact with a computer. Before the advent and popularity of graphical user interfaces (GUIs), the terminal and the shell were the only mechanism to make the computer do what we wanted it to do. At first glance, one may wonder what the shell does – other than passing commands to the underlying operating system for execution. Most of us are familiar with commands like ‘ls’ (for listing contents of a directory), ‘cd’ (for changing the current directory), and so on. It is through the shell that we can execute these commands. The shell understands the text we type — converts it into tokens — and then executes them on the operating system.

Flavours

Initially, the terminal started with the humble Bourne shell or ‘sh’. Over the years, many shell variants were developed and used. Some of the popular ones are ‘C Shell’ / ‘csh’ and ‘Korn Shell’ / ‘ksh’. ‘sh’ fell out of favour for a few years, but has gained popularity once again through its recent avatar, namely ‘bash’ / ‘Bourne Again Shell’.

What does the shell actually do?

The shell is the immediate interface between the operating system (OS) and the user. We make the computer do what we want, by using commands and applications supported by the tools installed on the computer we are using. Some commands are applications installed on the operating system, while some are built into the shell itself. Some of the commands built into bash are ‘clear’, ‘cd’, ‘eval’, and ‘exec’, to name a few, while commands like ‘ls’, and ‘mkdir’ are applications. The commands built into the shell vary as per the shell.

In this article, we cover a few aspects related to ‘bash’.

More about the shell

Most of us have used commands like ‘ls’, ‘cd’, and ‘mkdir’. When we run the ‘ls -l’ command on a directory, all the directories and files in that directory are listed on the screen.